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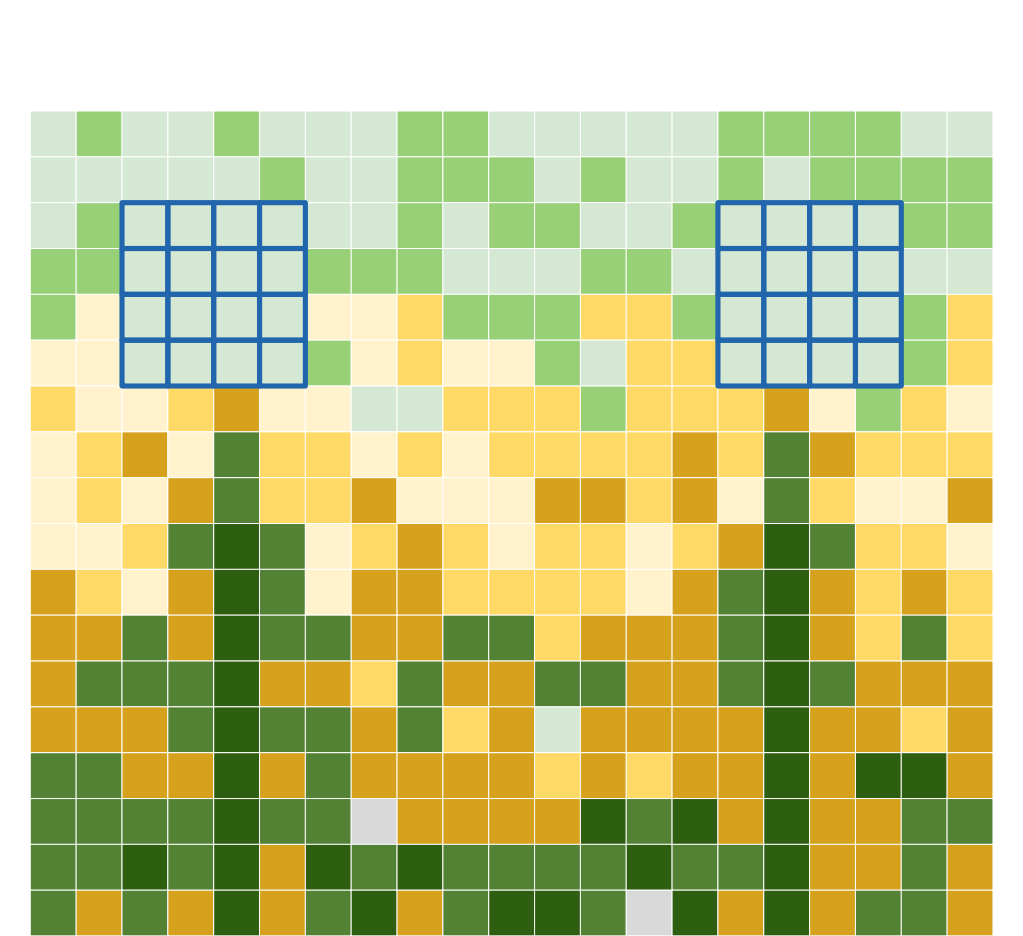
MA324 Project

Wildfire Firebreak Planning

1 Context

A regional fire authority manages a rural landscape in a fire-prone area. The landscape is represented as a 21×21 grid, where each cell contains one of four vegetation types: bare ground, grassland, shrubland, or forest. Each vegetated cell also has a density rating (sparse, normal, or dense). A small settlement occupies the north-east of the grid, and an ecologically sensitive wetland sits in the north-west.



Landscape map

Both settlement and wetland must be protected from wildfire. The authority estimates that fire damage to the settlement is more costly than the damage to the wetland: each burned settlement cell costs 10 units and each burned wetland cell costs 5 units.

During the fire season, lightning strikes occasionally ignite fires in the landscape. Once ignited, fire spreads to neighbouring cells with a probability that depends on vegetation type, vegetation density, and wind conditions.

The fire authority can clear firebreaks: remove vegetation from some cells, blocking fire propagation entirely. The fire authority has a budget of $B + R = 17$ cells that can be cleared. $B = 13$ cells can be permanently cleared before the fire season. On top, $R = 4$ cells can be cleared after wind conditions are revealed at the start of the fire season (before the fire actually ignites).

1.1 Data

You are provided with:

- `landscape.csv`: A vegetation and density map of the landscape. It is a 21×21 matrix encoding vegetation type and density for each cell, no header, comma-separated. Each entry encodes vegetation type and density as a two-digit integer: the tens digit gives vegetation (1=grassland, 2=shrubland, 3=forest) and the units digit gives density (1=sparse, 2=normal, 3=dense). Bare ground is coded as 0 (no vegetation or density). Cells that are already bare ground do not need clearing and do not cost budget. The landscape is written row-major (row 1 = top = north).
- `fire_simulator.r`: An R function that takes the landscape (with firebreaks marked as bare ground), an ignition cell, wind speed (km/h), and wind direction (radians), and returns which cells burned. The function documentation describes the spread probability formula used internally, including the spread probability $p_{\text{burn}}(i \to j)$ for each pair of adjacent cells.
- `targets.csv`: The locations of the settlement cells and the wetland cells with columns "row", "col", and "weight". Each row gives a protected cell's grid coordinates and its per-cell damage weight. The numbering of rows and columns uses the convention that row 1 = top = north.

The data and code are available on Moodle.

1.2 Uncertainty

Ignition location, wind speed, and wind direction are not known before the fire season: they are revealed at the start of the fire season. The fire authority's meteorological data suggests:

- **Wind speed** follows a Weibull distribution with shape $k = 2$ and scale $A = 7$ km/h, with density

  $$f_V(v) = \frac{k}{A}\left(\frac{v}{A}\right)^{k-1} \exp\left[-\left(\frac{v}{A}\right)^{k}\right], \quad v \geq 0.$$

  You may sample wind speeds using base R (no packages).

- **Wind direction** follows a mixture of two von Mises distributions: with probability 0.7 the wind blows from the west ($\mu_1 = 1.5\pi$, $\kappa_1 = 3$), and with probability 0.3 from the south-east ($\mu_2 = 0.75\pi$, $\kappa_2 = 2$). The mixture density is

  $$f_\Theta(\theta) = p_1 \cdot \frac{e^{\kappa_1 \cos(\theta - \mu_1)}}{2\pi\, I_0(\kappa_1)} + (1 - p_1) \cdot \frac{e^{\kappa_2 \cos(\theta - \mu_2)}}{2\pi\, I_0(\kappa_2)}, \quad \theta \in [0, 2\pi),$$

  where I_0 is the modified Bessel function of the first kind (available in R as `besselI(x, nu = 0)`). You will need this density to implement your sampler. For wind direction, you must implement your own sampler using acceptance-rejection (see Step 3). You may use the `circular` package (`rvonmises`) to validate your implementation, but not as your production sampler.

- **Ignition location** is uniformly distributed over all vegetated (non-bare) cells. You may sample ignition locations using base R (no packages).

1.3 Objective

The fire authority wants to minimise the total weighted damage to the settlement and wetland across fire scenarios. Since ignition, wind speed, and wind direction are random, you must decide how to measure performance under uncertainty: choosing an appropriate risk measure is part of the modelling task.

The authority prefers permanent plans that perform well even without reactive adjustments.

2 Current Solution

The authority currently maintains a single firebreak corridor along an existing track that runs east–west across row 10 (row 1 = north), using 17 cells in the centre and leaving 2 cells open on each side. They say: "This corridor has worked so far because it stops fires from the southern scrubland from reaching the northern settlement."

However, this corridor commits the entire combined budget to a single fixed barrier, leaving no capacity for reactive response once wind conditions are revealed.

3 Question

How should the authority allocate its firebreak budget to protect both the settlement and the wetland?

4 Scaffolding

Work through the following steps in order. Each step builds a model of increasing complexity: the first diagnoses where fire pressure concentrates, the second optimises firebreak placement under a fixed scenario, the third evaluates performance under uncertainty, and the fourth introduces reactive decision-making.

Each step produces distinct material for your report: a model, code, results, and a discussion addressing the questions posed. Work through as many steps as you can, then stop and write the report (see below for instructions on how to write your report).

4.1 Step 1: Network bottleneck model

4.1.1 Assumptions

- There is no wind.
- There is a fixed ignition: all southernmost cells start on fire (all cells in row 21).
- There is no budget limit: the goal is to find the minimum cost to fully protect each target (settlement and wetland).

4.1.2 Goal

Understanding where fire flows most strongly toward each target, if the fire were to start in the south.

4.1.3 Model

The grid defines a network: each cell is a node, and edges connect a cell with its adjacent cells (in all directions including diagonals). The simulator uses a spread probability $p_{\text{burn}}(i \to j)$ for each pair of adjacent cells, which depends on the vegetation type and density of the receiving cell (see the documentation). These spread probabilities can be interpreted as edge capacities: higher spread probability means the edge carries more "fire flow." The fire reachability from the ignition in the southernmost row to the settlement and the wetland can be modelled as a max-flow problem on this network.

1. Formulate and solve a max-flow problem in AMPL with the southernmost row as the source and the settlement cells as targets. Repeat for the wetland cells as targets.
2. Report the shadow prices on the edge capacity constraints. These identify **bottleneck corridors**: edges where reducing spread probability (by clearing a cell) would reduce fire flow the most.
3. Based on the bottleneck analysis, propose a firebreak plan: which cells would you clear to protect the settlement? Which ones to protect the wetland?

4.1.4 Discussion questions

- Does the current east–west corridor coincide with the bottleneck corridors your analysis identified? If not, what does the current corridor miss?
- Which target (settlement or wetland) is harder to protect? Why?
- This model treats spread probabilities as fixed capacities. What does it ignore?

4.2 Step 2: Budget-constrained firebreak optimisation

4.2.1 Assumptions

- There is no wind.
- There is a fixed ignition: all southernmost cells start on fire (all cells in row 21).
- The budget can be used only before the fire season starts.

4.2.2 Goal

Finding the optimal firebreak placement before the fire season starts, with limited budget and under deterministic conditions.

4.2.3 Model

Formulate a MILP that chooses which B cells to clear to minimise total weighted damage (settlement cells weighted 10 and wetland cells weighted 5). This model encodes fire propagation as constraints: a cell can burn only if it has a burning neighbour that could spread fire to it, and it has not been cleared.

The central modelling challenge is encoding stochastic fire spread as deterministic constraints. Two approaches are natural, and you should try both.

Model A: Binary propagation | Model B: Continuous reachability

An edge is "active" if $p_{\text{burn}} > \tau$ for some threshold $\tau \geq 0$. A cell burns if any burning neighbour has an active edge to it and the cell has not been cleared. This is a worst-case model: every active edge carries fire with certainty. The threshold τ controls how conservative the model is: lower τ includes weaker edges and requires more firebreaks.

Both models share the same budget constraint, clearing variables, and objective; they differ only in the propagation constraint. Choose several meaningful values of the budget B and proceed as follows for all the chosen values.

1. Solve both models in AMPL. Report the optimal firebreak locations and the resulting damage for each.
2. How does the budget affect the damage in each model?
3. Obtain the shadow price on the budget constraint. How much does one additional cleared cell reduce damage? Think carefully about whether the shadow price is well-defined for your model.
4. Compare the optimal plans from both models. Do they agree? If not, where do they disagree, and why?
5. Compare the optimal plan to the current east–west corridor and to the bottleneck-based plan from Step 1.
6. Run the fire simulator with your optimal firebreaks in place (no wind, southern front ignition). Does the simulated damage match either model's prediction? If not, what explains the gap? Also, simulate with no firebreaks to establish a baseline.

4.2.4 Discussion questions

- A model's *prediction* and the *plan it produces* are different things. A model can predict poorly but still produce a good plan (or vice versa). What does the simulator validation tell you about the quality of each model's predictions and plans?
- At what budget level does additional spending stop helping? Do the two models answer this question differently? What does this tell the authority about the minimum investment needed?
- The MILP finds the best plan for one specific ignition scenario and no wind. How likely is this plan to perform well under different conditions?
- If the damage weights on settlement and wetland cells were equal, would the plan change? What does this say about the implicit prioritisation in the model?

4.3 Step 3: Evaluation under uncertain wind

4.3.1 Assumptions

- Wind and ignition follow the distributions given in the Context.
- There is a single fire ignition, in a uniform random position.
- Only the budget $B = 13$ can be used, before the fire season starts.

4.3.2 Goal

Evaluating how the deterministic plan from Step 2 performs under realistic uncertainty, and whether it is better than the current solution.

4.3.3 Model

Select the firebreak plan from Steps 1 and 2 with budget $B = 13$ that you consider most appropriate and justify your choice. Evaluate it under uncertainty using Monte Carlo simulation.

1. Sample wind directions from the mixture distribution using **acceptance-rejection** with a suitable proposal distribution. Justify your choice of proposal distribution, report the acceptance rate, and discuss whether a better proposal could improve it. Verify that your sampler produces the correct density.
2. For each of N simulated fire scenarios (sampled ignition, wind speed, and wind direction), run the simulator with the firebreaks in place and record the damage. Choose N so that your qualitative conclusions correspond to the long-term behaviour of the values generated. Justify briefly your choice.
3. Report the expected damage, the $\mathrm{CVaR}_{0.9}$ of damage, and 95% confidence intervals for both.
4. Repeat the evaluation for the current east–west corridor and for no firebreaks at all. Compare the three plans using the risk measures from the previous point.

4.3.4 Discussion questions

- Under which wind conditions does the deterministic plan from Step 2 perform poorly? What characterises the worst 10% of scenarios? Consider both wind direction and ignition location.
- Your plan uses B cells while the corridor uses 17. What does the comparison of their risk profiles tell the authority about targeted placement versus broad coverage?
- The plan was optimised for no wind and south ignition. How far is it from what you would choose if you could see the wind in advance? You do not need to answer this precisely: a qualitative argument based on the scenarios where the plan fails is sufficient.
- Look at the distribution of damage across scenarios. Is expected damage or CVaR a better criterion for the fire authority's decision? What does the shape of the distribution tell you about each measure's usefulness?
- How many scenarios N are needed for the confidence intervals to be usefully narrow? What does this choice depend on?

4.4 Step 4: Reactive response

4.4.1 Assumptions

- Wind and ignition follow the distributions given in the Context.
- There is a single fire ignition, in a uniform random position.
- The budget can be used as indicated: $B = 13$ cells can be chosen before the fire season starts, and $R = 4$ cells can be chosen after wind conditions are revealed but before the fire ignites.

4.4.2 Goal

Quantifying the value of splitting the budget into permanent and reactive components, and identifying where reactive effort is most needed.

4.4.3 Model

The authority's decision has two stages: permanent firebreaks are committed before the season, and reactive firebreaks are chosen after wind is revealed but before ignition. The true ignition location and stochastic fire spread remain unknown at both stages. A fully optimal approach would jointly optimise permanent and reactive cells across all scenarios simultaneously: a two-stage stochastic program. Instead, we fix the permanent plan from Step 3 and use the deterministic reachability MILP (Step 2, Model B) to choose reactive cells for each wind realisation. A reactive MILP computes edge weights for the revealed wind. We will consider a full southern-front ignition as a conservative proxy. This is an approximation, and the discussion questions ask you to reflect on its limitations.

4.4.3.1 Overview

The analysis proceeds in three parts.

1. Use the permanent plan from Step 3 at budget $B = 13$. For each wind scenario from Step 3, solve a reactive MILP that adds at most $R = 4$ cells to the permanent firebreaks, using wind-dependent edge weights.
2. Solve the Step 2 MILP at budget $B + R = 17$ with no wind to obtain a permanent-only plan that uses the full combined budget. This represents committing all resources before the season.
3. Evaluate all plans under the same N scenarios using the simulator and compare their risk profiles.

4.4.3.2 Planning

Consider the solution you evaluated in Step 3. For each wind scenario, solve a budget-constrained MILP that adds at most $R = 4$ reactively cleared cells to the $B = 13$ permanent firebreaks. The permanent cells are fixed; only the reactive cells are decision variables. Recompute spread probabilities for the revealed wind speed and direction, then compress as in Step 2.

Each scenario requires a separate MILP solve, so total computation time scales with N and with the solver time limit per solve. You will need to balance solution quality (tighter solver tolerance), sample size (N), and total runtime. Start with small N and short time limits during development; increase for your final results. Report your choices and their consequences.

4.4.3.3 Evaluation

For each scenario, apply the permanent and reactive firebreaks to the landscape and run the simulator with the scenario's ignition, wind speed, and wind direction. Record the damage.

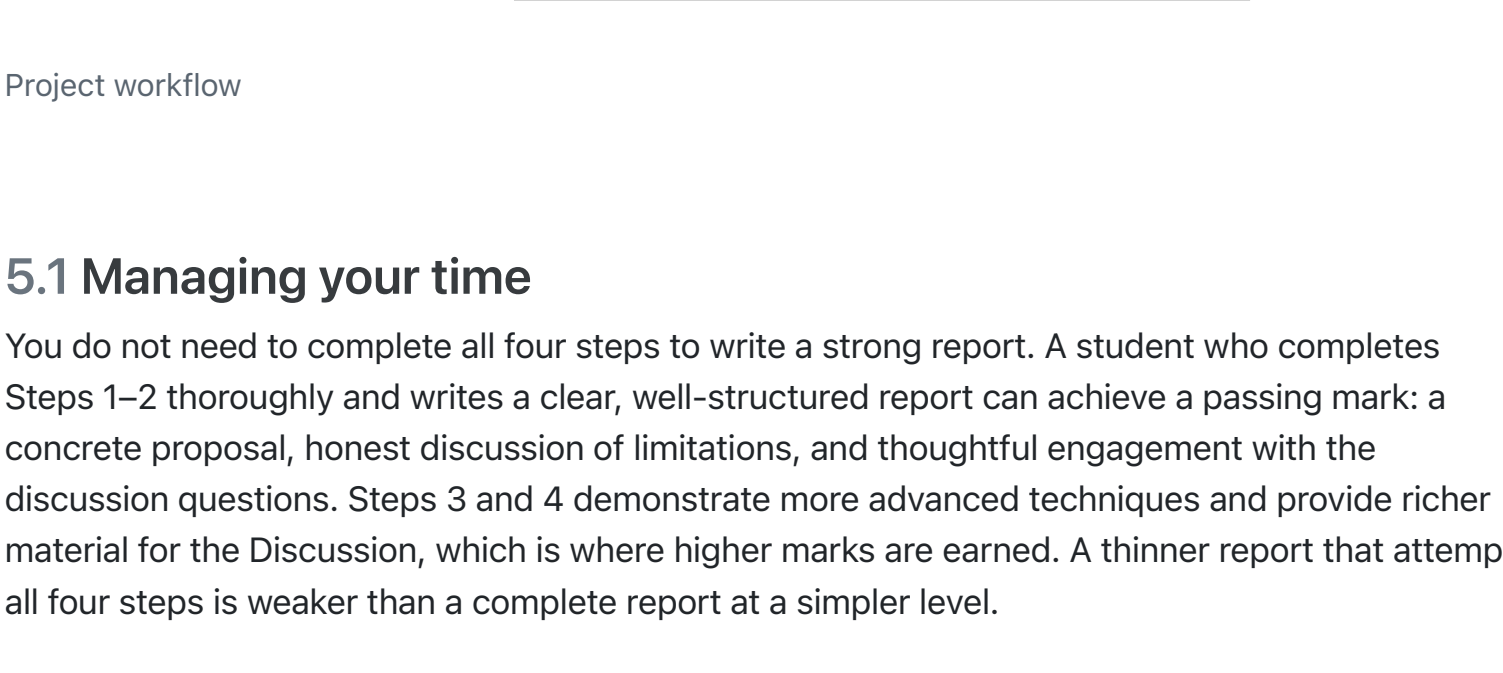
1. Solve the reactive MILP for each scenario. Record the damage and which cells are reactively chosen.
2. Evaluate four plans under the same N scenarios and compare their risk profiles (expected damage, $\mathrm{CVaR}_{0.9}$, confidence intervals):
   1. No firebreaks.
   2. $B = 13$ permanent cells only, i.e., Step 3 plan without reactive adjustment.
   3. $B + R = 17$ permanent cells, planned before the season using the Step 2 MILP with no wind.
   4. $B = 13$ permanent cells plus $R = 4$ reactive cells chosen per scenario as described above.
3. For each scenario, the **reactive effort gap** is the difference in damage between plans (2) and (4). Report the distribution of this gap. In which scenarios is reactive effort most valuable? What characterises those scenarios?
4. Explore the reactive budget R: how does increasing R affect the risk profile? At what point does additional reactive capacity stop helping?

4.4.4 Discussion questions

- The reactive MILP uses a deterministic reachability model to choose cells for each wind realisation. What would a fully optimal approach look like? What additional information would it use, and why might it be impractical?
- Compare plans (3) and (4): both use $B + R$ cells in total, but (3) commits all of them before the season while (4) reserves R for use after wind is revealed. In theory, (4) should perform at least as well as (3) because it uses the same budget with strictly more information. Do your results confirm this? If yes, which results confirm this? If not, which approximations in the reactive approach could explain the gap?
- The reactive MILP may not solve to optimality within your time limit. How does the solver's optimality gap affect the quality of the reactive plans?
- Which cells are most frequently selected as reactive firebreaks across scenarios? What does this spatial pattern reveal about the limitations of the permanent plan?
- The authority assigns weights to settlement and wetland differently. Does the pattern of reactive cell selection reflect this asymmetry?
- The reactive MILP fixes ignition as a southern-front proxy, so the only randomness in the planning phase is wind (speed and direction). Could you replace Monte Carlo sampling with a deterministic approach for this phase? What would this gain and what would it lose?
- The permanent and reactive budgets are fixed at $B = 13$ and $R = 4$. What would happen with $B = 12$ and $R = 5$? Re-solve Step 2 at the new B and repeat the analysis. How do the risk profiles change?

5 How to work through the project

Start with the scaffolding, where the analysis happens. Once you have enough results, write the Discussion: it draws on the step-level discussions but focuses on the bigger picture and connects technical findings to practical recommendations. Then write the Proposal: a concrete recommendation grounded in your results. Then, write the Current Solution section. This section introduces the problem from the authority's perspective, demonstrates your understanding of their reasoning, and sets up the issues that your proposal addresses. It is the first section the reader sees, so it should be written with the full picture in mind. Finally, write the AI declaration giving complete account of your use of AI during the project.



Project workflow

5.1 Managing your time

You do not need to complete all four steps to write a strong report. A student who completes Steps 1–2 thoroughly and writes a clear, well-structured report can achieve a passing mark: a concrete proposal, honest discussion of limitations, and thoughtful engagement with the discussion questions. Steps 3 and 4 demonstrate more advanced techniques and provide richer material for the Discussion, which is where higher marks are earned. A thinner report that attempts all four steps is weaker than a complete report at a simpler level.

5.2 Advice

- Work through the scaffolding until you feel you have enough material for a complete report. During development, to check preliminary results and your code, run fewer iterations and stop optimisation solvers early.
- **Stop and write the report.** Writing the report requires a complete overview and cannot be done sequentially. You can always return to the scaffolding if time permits.

6 What to deliver

Submit your report on Cadmus by the deadline. Your report must include all analysis code directly in the document. There is no separate file submission.

6.1 Report structure

Your report has the following structure:

1. **Current Solution.** A critique of the authority's current solution, noting strengths and weaknesses.
2. **Proposal.** Your recommended firebreak plan.
3. **Scaffolding.** Each step you completed, with subsections for Model, Code, Results, and Discussion.
4. **Discussion.** Synthesis across steps.
5. **AI Declaration.** How you used AI tools and what parts are your own work.

6.2 Code

R code longer than 10 lines must be implemented as R functions and documented using roxygen2-style comments (#' with @param, @return) as introduced in Workshop 2 (available on Moodle). AMPL code should include comments explaining any non-obvious logic.

6.3 Tables and figures

Tables must not exceed 6 rows and 6 columns (excluding headers): larger comparisons must be presented as figures. Figures should have informative titles that state the finding, not just describe the chart type, as introduced in Workshop 4 (available on Moodle).

6.4 Word limit

The word limit depends on how many steps you include in your report. All text counts towards the limit, including code. Counting code documentation towards the word limit is part of the assignment. Figures and plots (images) do not count. You do not need to include the code that generates your images: the figures themselves demonstrate your visualisation choices.

ⓘ Word limits

Steps included	Word limit
Steps 1–2	4000
Steps 1–3	5000
Steps 1–4	7000

This is automatically counted in Cadmus for your reference. Embedding text or code as images to circumvent the word limit will be harshly penalised. Concise and focused writing is part of the assignment.

6.5 Reproducibility

Use `set.seed(1)` at the beginning of your code. This ensures your results are reproducible. Moreover, your results should not qualitatively change if you change the seed, so make sure this is the case.

6.6 Solvers

You may use any solver available through your AMPL licence.

7 Marking scheme

7 Marking scheme

7.1 Report

The report will be marked as follows.

ⓘ **Marking scheme**

Category	Weight	Description
Modelling	20%	Formulations are correct and clearly stated. Modelling choices are justified. Trade-offs between models are understood.
Implementation	10%	Code is correct, readable, and reproducible. AMPL models are clean and data is separated from the model.
Analysis and Discussion	40%	Step-level discussions address the posed questions with specific evidence. The top-level Discussion synthesises findings, acknowledges limitations honestly, and connects technical results to practical recommendations.
Communication	20%	The report is well-structured and clearly written. The Current Solution section demonstrates understanding. The Proposal is concrete and grounded in results. Figures and tables are informative.
AI Usage	10%	The AI Declaration is honest, specific, and reflective. It distinguishes clearly between what was delegated, co-created, and produced independently.

7.2 Oral defence

Selected students may be invited to an oral defence where they will be asked to explain and justify their work. The oral defence may adjust the final mark in any direction.